Numeric versions of date and time

- In SDTM datasets,
    - the date and time values are stored in character variables
    - dates usually correspond to a collection or observation date, start date and end date of an event or an intervention etc
    - Start dates have a prefix of ST, and end dates have a prefix of EN (eg: aeSTdtc, aeENdtc, exSTdtc, exENdtc etc)
    - Collection or observation dates do not have a specific prefix other than the domain name (eg: dmDTC, vsDTC, lbDTC etc)

- For analysis purposes, we may want to separate the date and time components and create numeric versions of them
- ADaM standard allows us to create numeric versions of the date and time values
- Similar to other analysis variables, these numeric date/time/datetime variables will have a prefix of 'A' (begin with 'A') - 'A' stands for 'Analysis'.
- Similar to SDTM, start is abbreviated as 'ST', end is abbreviated as 'EN'
- Date variables have a suffix of 'DT', time variables have a suffix of 'TM' and datetime variables have a suffix of 'DTM'.
- For any SDTM date variable, 3 numeric counterparts can be created in ADaM based on the precision of information collected. Below is an example using AESTDTC.
    - ASTDT - to represent date component of AESTDTC in numeric format
    - ASTTM - to represent time component of AESTDTC in numeric format
    - ASTDTM - to represent the datetime value of AESTDTC in numeric format
- Similary, AENDT, AENTM, AENDTM are used to store the end date information
- For collection or observation dates like VSDTC, LBDTC we can use
    - ADT for date component
    - ATM for time component
    - ADTM for datetime value

Imputations

- In SDTM datasets, we just orgainze the collected date time values in ISO 8601 format but do not change them
- This means, if a subject could recollect only the year and month of an adverse event date, we just report the year and month in AESTDTC in AE domain
- For analysis purposes, we may want to 'assume' a value for the missing date or time components and use that value when creating analysis version date, time or datetime variables
- These imputation rules must be/generally pre-specified in the protocol or statistical analysis plan of the study
- ADaM standard allows for these imputations
- When the date, time or datetime value on a particular record is imputed, we want to clearly communicate that imputation is performed
- -DTF, -TMF varibales are used to communicate the level of imputatoin performed on a particular variable of a record
- The allowed values in -DTF variable are
    - Y - to indicate that year is imputed
    - M - to indicate that month is imputed
    - D - to indicate that day is imputed
    - note, when we impute multiple components we use the letter corresponding to highest level of imputation. For example, when month and day are imputed on a record, we populate the variable with 'M' as the highest level of imputation is at month. Similarly, when year and day are imputed, we populate it 'Y' as the highest level of imputation is year here.
- The allowed values in -TMF variable are
    - H- to indicate that hour is imputed
    - M - to indicate that minute is imputed
    - S - to indicate that second is imputed